KTH Royal Institute of Technology

Project Carrier Course for Medical Engineers, Part 1

# VibraTrack

## Real-Time Hand–Arm Vibration Monitoring and Exposure Estimation Using Wearable Sensors

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## Abstract

TBC.

## Acknowledgements

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# List of Acronyms

**BLE** Bluetooth Low Energy. 5–8, 11, 12

**HAV** Hand-Arm Vibration. 2

**HAVS** Hand-Arm Vibration Syndrome. 1, 2

**RMS** Root Mean Square. 7, 9–13

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# 1 Introduction

Vibrating handheld power tools such as drills, grinders, and sanders are widely used in occupational environments. Prolonged exposure to mechanical vibrations transmitted through the hand–arm system can lead to adverse health effects, including vascular, neurological, and musculoskeletal disorders. One of the most well-documented consequences of such exposure is Hand-Arm Vibration Syndrome (HAVS), a condition characterized by pain, numbness, reduced tactile sensitivity, and long-term loss of hand function [1, 2]. To mitigate these risks, international standards such as ISO 5349 define procedures for assessing vibration exposure and estimating daily exposure metrics such as the $A(8)$ value [3].

In practice, vibration exposure assessment is typically performed using accelerometers mounted directly on the tool handle and evaluated under standardized or laboratory conditions. While these approaches provide reproducible and comparable measurements, they are primarily intended for compliance testing and are not well suited for continuous monitoring during real working activities [4, 5]. As a result, workers generally lack practical feedback on their vibration exposure during normal tool operation.

Wearable inertial sensors placed on the hand or wrist have emerged as a promising alternative, enabling measurements closer to the point where vibration enters the body. Previous studies have shown that different power tools generate characteristic vibration patterns across a broad frequency range, typically spanning from approximately 20 Hz to several hundred hertz [6]. This suggests that a single hand-mounted sensor, combined with appropriate signal processing, could be used to detect periods of vibration exposure in realistic environments.

## 1.1 Aim of the Project

The **goal** of this project is to investigate the feasibility of using a hand-mounted wearable inertial sensor combined with real-time mobile signal processing to detect periods of hand–arm vibration exposure. Specifically, the work aims to design and implement a lightweight, on-device processing pipeline capable of distinguishing between vibration exposure and non-exposure during the use of handheld power tools, while operating under the computational and latency constraints of a mobile application. Activities involving a drill, grinder, sander, and multifunctional tool are used as development cases to support exploratory method design and validation [1, 6].

# 2 Background

This chapter establishes the theoretical and technical foundations of the project. It introduces Hand-Arm Vibration (HAV) exposure and its associated health risks, reviews standard measurement approaches and recent developments in wearable sensing, and discusses signal characteristics and engineering constraints that motivate lightweight real-time vibration monitoring strategies.

## 2.1 Hand–Arm Vibration Exposure (HAV)

HAV refers to mechanical vibrations transmitted from handheld power tools to the human hand and arm. Prolonged exposure represents a well-established occupational risk and can lead to permanent impairments affecting blood circulation, peripheral nerves, and muscles [1]. These conditions are collectively referred to as HAVS [2].

The severity of HAV-related health effects depends on vibration amplitude, frequency content, exposure duration, and coupling conditions between the tool and the hand. To manage these risks, international standards require vibration exposure to be quantified. Exposure is commonly expressed as an equivalent daily vibration value normalized to an 8-hour working period, known as $A(8)$ [3].

## 2.2 Wearable Sensors for Vibration Monitoring

Standardized assessment of hand–arm vibration exposure is traditionally performed using accelerometers rigidly mounted on power tool handles [3]. While this approach accurately characterizes tool-emitted vibration, it does not fully account for how vibration is transmitted to the worker, which depends on factors such as grip force, posture, and protective equipment [2].

Wearable sensors placed on the hand or wrist enable measurement closer to the user's actual exposure and under realistic working conditions [5]. However, wearable measurements introduce challenges, including signal attenuation due to soft tissue and gloves, increased sensitivity to motion artifacts, and reduced comparability with standardized tool-mounted measurements [4].

## 2.3 Previous Work and Scientific Context

Prior research has shown that vibration amplitudes measured on the hand or wrist are generally lower than those measured directly on tool handles due to the biodynamic response of the hand–arm system [2]. Several studies have proposed transfer functions to relate wearable measurements to ISO-standard references. For example, Mootz and Akhavian [7] demonstrated that calibrated wearable measurements can approximate standardized exposure estimates.

Other work has focused on spectral analysis of vibration signals. Matthies et al. [6] showed that frequency-domain features extracted from wearable sensors can distinguish between different power tools, highlighting the relevance of spectral descriptors for vibration monitoring.

## 2.4 ISO Standards as Reference Framework

ISO 5349-1 [3] defines standardized procedures for measuring and evaluating human exposure to hand-transmitted vibration. The standard specifies tri-axial acceleration measurement and combines the axes using a root-sum-of-squares formulation.

The standard further accounts for frequency-dependent health effects through weighting curves.

While essential for compliance testing, direct implementation of full ISO weighting is challenging in real-time wearable systems. In wearable monitoring scenarios, exposure estimation is therefore often approximated by focusing on signal energy within tool-relevant frequency bands rather than by implementing full ISO weighting and calibration procedures. Such approximations are commonly adopted in exploratory and feasibility studies, while acknowledging their limitations with respect to standardized exposure assessment, as we intended to do.

### 2.5 Characteristics of Vibration Signals

Human motion primarily generates low-frequency signal components, typically below 20 Hz. In contrast, powered hand tools produce vibration signals with substantial energy at higher frequencies, often extending into the several-hundred-hertz range [1].

Tool-induced vibration also tends to be more stationary than voluntary human motion, producing sustained signal segments during operation [8]. These characteristics motivate the use of band-pass filtering and short-time windowing for vibration detection.

### 2.6 Motivation for Simple Vibration Classification

Real-time wearable monitoring systems must operate continuously under strict computational and energy constraints. Although advanced machine learning methods can provide detailed classification, they often require substantial processing resources and are less suitable for long-duration on-device execution [9].

By formulating vibration detection as a binary classification problem, the system can operate efficiently and robustly in real time. The use of simple, interpretable signal features enables reliable detection of exposure periods while prioritizing accurate estimation of exposure duration.

# 3 Materials

## 3.1 Hardware Setting

**Tools** For this project, four different handheld power tools were selected, with the intention of capturing vibration signals spanning a wide range of dominant frequency bands, typically encountered in occupational hand-arm vibration exposure. The selection was driven by the need to include tools characterized by low-, medium-, and high-frequency vibration components [1].

The tools used are summarized as follows:

- **Orbital sander (PSM)**: characterized by medium-frequency vibrations, with dominant oscillation components typically observed in the range of approximately 100–250 Hz.
- **Electric drill**: mainly producing low to medium frequency vibrations, generally between 10–100 Hz. Under load conditions, pronounced low-frequency components are commonly observed.
- **Angle grinder (PLSH)**: dominated by high-frequency vibration components, often exceeding 300 Hz during operation.
- **Multifunction oscillating tool (PMF)**: producing high-frequency vibrations, typically in the range of several hundred Hertz depending on the selected accessory.

It is important to note that the effective vibration spectrum generated by each tool is not fixed. The dominant frequency bands may vary depending on several factors, including the accessory used, the applied force or load during operation, the mounting location of the sensor, and the operator's grip and contact material [2, 3].




(a) Orbital sander (PSM)




(b) Electric drill



(c) Angle grinder (PLSH)



(d) Multifunction tool (PMF)

Figure 1: Handheld power tools used for vibration data collection: orbital sander (PSM), electric drill, angle grinder (PLSH), and multifunction oscillating tool (PMF).

**Wearable Sensor: Movesense** Vibration data were collected using a **Movesense wearable sensor**, equipped with a triaxial accelerometer capable of measuring linear acceleration along three orthogonal axes [9]. The sensor provides raw acceleration measurements and supports configurable sampling frequencies suitable for capturing both low- and high-frequency vibration components.

As shown in Figure 2, the sensor was mounted on the dorsal (upper) side of the operator's (one group member) hand and secured tightly using a glove. This placement was chosen to capture vibrations transmitted through the hand–arm system during tool operation, providing a measurement closer to user-relevant exposure than rigid tool-mounted measurements [5]. Since the sensor is not in direct contact with the tool housing, the recorded signal reflects attenuation and spectral shaping introduced by the hand–glove coupling; consequently, absolute vibration amplitudes cannot be directly interpreted as tool-source vibration, and placement/coupling, also considering motion artifacts (hand movement), may contribute to increased variability [7].

The Movesense sensor communicates wirelessly with the mobile application via Bluetooth Low Energy (BLE), enabling continuous data streaming during tool usage. Acceleration data are transmitted in a structured digital format, including timestamps and axis-specific measurements, allowing synchronization with downstream processing pipelines.




Figure 2: Movesense wearable sensor used for triaxial acceleration measurement and typical placement on the operator's hand during tool operation.

## 3.2 Software & Development Environment

The software infrastructure for this project consists of an offline development environment and a mobile application environment. This subsection describes the software tools and execution contexts used to develop, test, and support the proposed system. Algorithmic details of signal processing and classification are described separately in the Methods section.

**Offline Development Environment** Offline development and analysis were conducted using the Python programming language. Python was selected due to its wide adoption in scientific computing and its mature ecosystem for numerical analysis and signal inspection. The offline environment was used to inspect raw sensor data, develop and test classification approaches, and generate intermediate and final datasets for evaluation.

Raw acceleration data acquired from the Movesense sensor were stored in structured JSON files and organized according to tool type and recording session. Processed data and derived representations were exported in CSV and JSON formats to facilitate inspection, visualization,

and reproducibility. The use of standardized file formats also enables future extensions of the system, including the application of alternative and more advanced classification approaches.

**Mobile Application Development** The mobile application was developed for the iOS platform using Swift and SwiftUI within the Xcode development environment. The application provides the execution context for receiving acceleration data from the Movesense sensor via BLE and serves as the foundation for future possible real-time deployment of the proposed vibration detection logic.

### 3.3 Dataset

The dataset used in this project consists of triaxial acceleration recordings collected using the Movesense wearable sensor during the operation of handheld power tools. Data were acquired across multiple recording sessions and group members under controlled conditions.

Each recording contains time-stamped acceleration samples along three orthogonal axes. Recordings were organized by tool type and operational condition and stored in structured JSON files. In addition to vibration recordings, non-vibration segments corresponding to background activities were also collected to support the development and validation of vibration detection and classification logic.

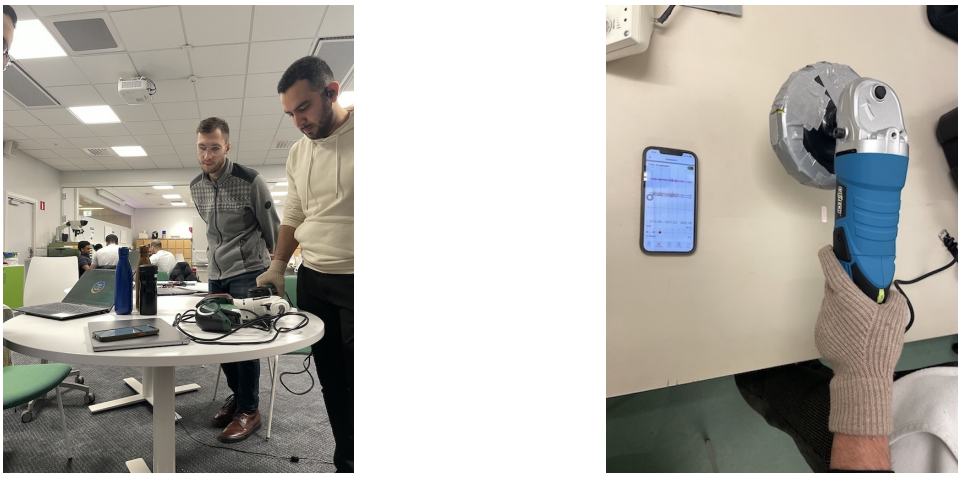


Figure 3: Experimental recording setup used for vibration data collection.
**Left:** laboratory setup.
**Right:** mobile application interface during active triaxial acceleration recording.

# 4 Methods

## 4.1 System Overview

The proposed vibration detection system is implemented as a fully real-time mobile application that processes streaming acceleration data acquired from a wearable sensor. The system follows a structured signal processing pipeline that transforms raw tri-axial acceleration measurements into discrete vibration activity states and cumulative exposure indicators. An overview of the implemented real-time pipeline is shown in Figure 4.

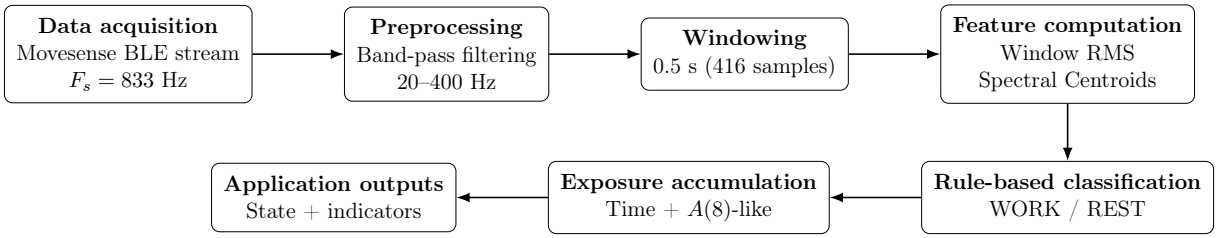



Figure 4: Real-time signal processing and classification pipeline implemented in the mobile application. The pipeline operates sequentially on streaming acceleration data acquired via BLE.

Table 1: Real-time processing parameters used in the mobile application.

| Step | Setting |
|---|---|
| Sampling frequency | 833 Hz (Movesense BLE stream) |
| High-pass filter | 20 Hz (IIR, cascaded second-order sections) |
| Low-pass filter | 400 Hz (IIR, second-order) |
| Window length | 0.5 s (416 samples) |
| Window overlap | None (sequential windows) |

Raw tri-axial acceleration data are acquired directly from the Movesense wearable sensor via BLE at a fixed sampling frequency of 833 Hz. Incoming samples are processed sequentially as they are received, without intermediate file storage or batch-level operations.

The streaming acceleration signals undergo a lightweight preprocessing stage consisting of band-pass filtering between 20 and 400 Hz. This filtering stage is designed to suppress quasi-static components and low-frequency motion artifacts, as well as attenuate high-frequency noise, while preserving frequency components typically associated with hand–arm vibrations generated by power tools. The selection of cutoff frequencies and filter structure was informed by exploratory time-domain and spectral analyses performed on representative recordings from different tools (Appendix B).

Filtered samples are aggregated into fixed-length, non-overlapping windows corresponding to approximately 0.5 s of data. For each window, compact time- and frequency-domain descriptors are computed, including the Root Mean Square (RMS) vibration magnitude and the spectral centroid. These features are subsequently compared against a predefined threshold to classify each window as either `WORK` (vibration exposure) or `REST`.

When a window is classified as `WORK`, the system increments cumulative exposure time and updates exposure-related indicators, including a running $A(8)$-like metric normalized to an 8-hour reference duration, used here as a relative exposure proxy. Classification decisions and exposure metrics are propagated to the user interface in real time, enabling continuous feedback during tool operation.

Furthermore, during system development, additional exploratory signal analyses were conducted to guide the selection of preprocessing parameters and window lengths; these analyses are reported in Appendix B.

## 4.2 Data Acquisition

Acceleration data acquisition was performed using the Movesense wearable sensor described in the previous section, with the mobile application operating in a fully real-time streaming configuration. In the final system setup, the application subscribes directly to the accelerometer data stream via BLE using the Movesense MDS interface.

The accelerometer stream was acquired at a fixed sampling frequency of 833 Hz by subscribing to the dedicated MDS endpoint `/Meas/Acc/833`. This sampling rate was selected to ensure sufficient temporal resolution for capturing vibration components associated with handheld power tools, which may extend into the several-hundred-hertz range. The sampling frequency is defined by the selected subscription endpoint and does not require estimation or correction during acquisition.

Incoming data packets contain an array of tri-axial acceleration samples (`ArrayAcc`), with each sample consisting of instantaneous acceleration values along the $x$, $y$, and $z$ axes. Samples are parsed sequentially upon reception and immediately forwarded to the processing pipeline. No intermediate file storage or batch-level buffering is performed in the real-time configuration.

Data acquisition proceeds continuously during tool operation, without explicit segmentation at the acquisition stage. An example of the raw acceleration signal acquired during tool operation is shown in Figure 5, where the characteristics of the raw acceleration signal motivate the preprocessing steps described in the following subsection.

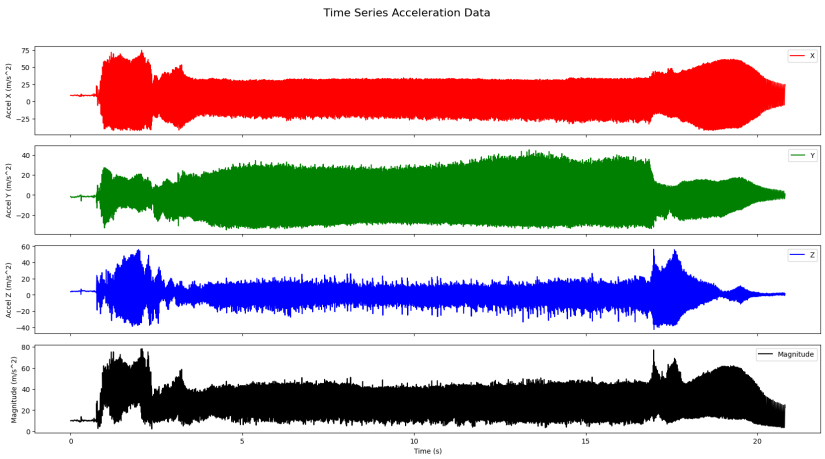



Figure 5: Representative example of raw tri-axial acceleration data acquired via BLE during operation of a handheld angle grinder. The three upper panels show the unprocessed acceleration components along the $x$, $y$, and $z$ axes, while the lower panel shows the corresponding acceleration magnitude. The plot illustrates the high-frequency oscillatory content and amplitude variability of the signal prior to any preprocessing, filtering, or segmentation.

## 4.3 Preprocessing

In the final system configuration, preprocessing is performed directly on the streaming tri-axial acceleration data in real time, prior to window-level feature computation. The goal of this stage

is to attenuate non-informative signal components (e.g. quasi-static offsets and slow hand-motion artifacts) while preserving the vibration band typically associated with handheld power tools.

Preprocessing consists of an axis-wise band-pass filtering stage with cutoffs at 20 Hz and 400 Hz. The filtering is implemented as a computationally lightweight IIR cascade: a 20 Hz high-pass section applied twice in series (to obtain a steeper low-frequency attenuation), followed by a 400 Hz low-pass section. This design enables sample-by-sample processing with low latency and modest computational cost, which is required for continuous on-device execution.

From a signal-processing perspective, each acceleration axis is filtered independently using a causal discrete-time IIR filter, which can be expressed as

$$y[n] = \sum_{k=0}^{M} b_k \, x[n-k] - \sum_{k=1}^{N} a_k \, y[n-k], \tag{1}$$

where $x[n]$ and $y[n]$ denote the input and output samples, respectively, and $\{a_k, b_k\}$ are the filter coefficients. The overall band-pass behavior is obtained by cascading high-pass and low-pass stages, such that

$$H_{\mathrm{BP}}(z) = H_{\mathrm{HP}}(z)\, H_{\mathrm{LP}}(z), \tag{2}$$

with cutoff frequencies at 20 Hz and 400 Hz. This configuration suppresses low-frequency motion components while retaining the frequency range in which tool-induced vibrations predominantly occur [8, 10].

The choice of IIR filtering reflects practical constraints of a mobile real-time pipeline. In particular, zero-phase forward–backward filtering (e.g. `filtfilt`) is not applicable in streaming operation because it requires access to future samples and introduces non-causal behavior. In our implementation, filtering is therefore causal and executed sequentially as samples are received. Filtered samples are immediately forwarded to the windowing stage for short-time analysis.

*Note on offline prototyping.* During early offline development, alternative preprocessing configurations were explored (e.g. lower high-pass cutoffs and zero-phase filtering) to support signal inspection and dataset preparation. The final real-time pipeline adopts the simplified 20–400 Hz causal band-pass as a pragmatic compromise between signal selectivity and real-time limitations, while remaining consistent with established practices in vibration signal analysis [2].

### 4.4 Feature Extraction

Feature extraction is performed on short, fixed-length windows of preprocessed acceleration data and aims to summarize the vibration content of each window using a compact and interpretable set of descriptors. In the final system configuration, feature extraction is intentionally kept minimal in order to support robust real-time execution on a mobile device.

Filtered tri-axial acceleration samples are aggregated into non-overlapping windows of approximately 0.5 s duration (416 samples at 833 Hz). For each window, two complementary scalar features are computed: the RMS vibration magnitude and the spectral centroid (Figure 6).

The RMS value provides a measure of signal energy over the window and is widely used in vibration analysis and exposure assessment due to its physical interpretability. Given the filtered acceleration components $x[n]$, $y[n]$, and $z[n]$ within a window of length $N$, the window-level RMS vibration magnitude is computed as

$$a_{\mathrm{RMS}} = \sqrt{\frac{1}{N} \sum_{n=1}^{N} \left(x^2[n] + y^2[n] + z^2[n]\right)}. \tag{3}$$

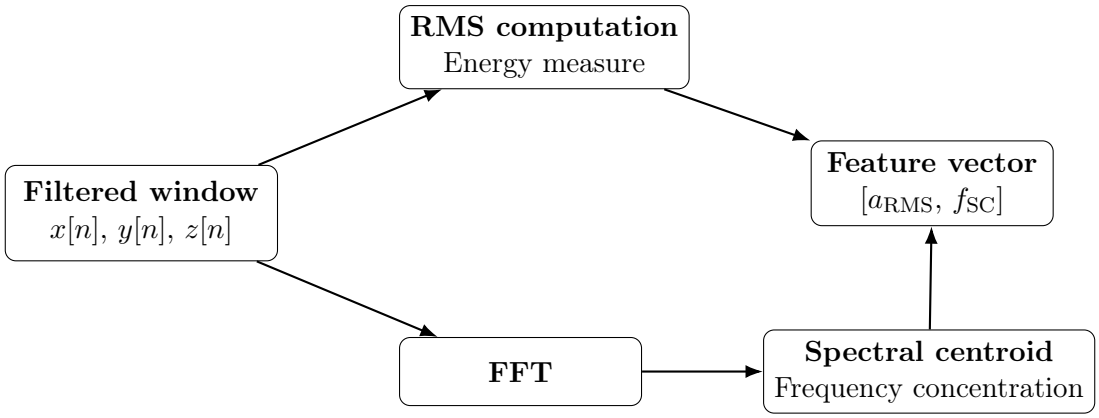



Figure 6: Schematic overview of window-level feature extraction. For each fixed-length window of filtered tri-axial acceleration data, an RMS vibration magnitude and a spectral centroid are computed, yielding a compact feature representation used for subsequent classification.

This formulation implicitly combines the three orthogonal acceleration components into a single vibration metric, capturing the overall vibration intensity experienced by the hand within the window. The use of RMS aggregation attenuates the influence of short impulsive fluctuations while preserving sustained vibration activity, making it suitable for exposure estimation and our threshold-based classification.

In addition to RMS, the spectral centroid is computed to capture information about the distribution of signal energy across frequencies. The spectral centroid provides a compact descriptor of the "center of mass" of the magnitude spectrum and is commonly used to distinguish between signals dominated by low- versus high-frequency components. For a given window, it is defined as

$$f_{\mathrm{SC}} = \frac{\sum_k f_k \, |X[k]|}{\sum_k |X[k]|}, \tag{4}$$

where $X[k]$ denotes the discrete Fourier transform of the windowed signal and $f_k$ the corresponding frequency bins.

The inclusion of the spectral centroid was motivated by exploratory offline analyses, which indicated that different tools exhibit distinct frequency concentration patterns even when RMS levels are comparable (Appendix B). While the centroid is not intended as a full spectral characterization, it provides a frequency-domain cue that complements RMS without substantially increasing computational complexity.

More complex time–frequency descriptors were considered during early experimentation but were not retained in the final real-time pipeline; illustrative examples are provided in Appendix B.

### 4.5 Rule-Based Classification

Window-level vibration activity is determined using a light rule-based classifier operating directly on the extracted features. The objective of this stage is to provide a robust and computationally efficient discrimination between periods of vibration exposure and non-exposed states under real-time constraints (binary classification task).

For each 0.5 s window, the classifier evaluates the RMS vibration magnitude $a_{\mathrm{RMS}}$ and the spectral centroid $f_{\mathrm{SC}}$ against predefined thresholds. A window is classified as `WORK` (vibration exposure) if both of the following conditions are satisfied:

$$a_{\mathrm{RMS}} > \theta_{\mathrm{RMS}}, \qquad f_{\mathrm{SC}} > \theta_{\mathrm{SC}}, \tag{5}$$

where $\theta_{\text{RMS}}$ and $\theta_{\text{SC}}$ denote empirically selected thresholds for vibration intensity and frequency concentration, respectively. Windows not satisfying these criteria are labeled as `REST`.

The RMS threshold enforces a minimum vibration energy level, effectively suppressing low-amplitude signals arising from hand motion, handling, or environmental noise. The spectral centroid threshold complements this criterion by requiring that a sufficient proportion of the signal energy is concentrated in higher frequency bands typically associated with powered tool operation. This dual-condition strategy improves robustness in scenarios where transient movements may produce elevated RMS values without sustained high-frequency vibration content.

Threshold values were determined through exploratory offline analysis of recorded signals from different tools and non-tool activities (Appendix B). This analysis was used to guide the selection of empirically meaningful operating points for the classifier, without resorting to formal machine-learning optimization.

This choice offers full transparency, minimal computational cost, and deterministic behavior, making it well suited for continuous on-device execution. Furthermore, the window-level decisions integrate naturally with exposure accumulation logic, enabling straightforward computation of cumulative exposure metrics such as time-above-threshold and the $A(8)$-like indicator described in the previous section.

### 4.6 App Integration Logic

The mobile component of the project was implemented in Swift and evolved through two complementary execution modes: (i) an *offline analysis mode* based on JSON file import (SwiftUI), developed in the early stages of the project, and (ii) a *real-time prototype mode* based on direct BLE streaming from the Movesense sensor (UIKit), which constitutes the final application configuration. This staged design approach was intentionally adopted to balance rapid development and validation with the investigation of real-time feasibility. The offline mode supported controlled inspection and reporting of recorded data during development, while the real-time mode provides a lightweight on-device implementation of vibration detection under streaming constraints.

**Offline analysis mode (SwiftUI JSON importer)** In an initial development phase, the application supported an offline analysis mode in which acceleration recordings stored as JSON files produced by the Movesense Showcase app could be imported and analyzed. The JSON structure was parsed using a dedicated `Codable` data model that mirrors the native sensor payload format, consisting of a root `data` list in which each entry contains an `acc` object with a millisecond-resolution timestamp and an optional `ArrayAcc` field holding tri-axial acceleration samples.

This mode was primarily intended for post-hoc analysis, visualization, and validation of vibration exposure patterns during algorithm development. For each imported recording, the application computed a summary report including total recording duration, maximum observed vibration amplitude, cumulative time spent above a predefined safety threshold, and a log of high-vibration events. Vibration magnitude was computed sample-wise as the Euclidean norm of the tri-axial acceleration vector. A simple gravity compensation was applied by subtracting the gravitational constant $g$ from the magnitude, yielding an estimate of the dynamic acceleration component.

High-vibration events were defined as contiguous time intervals during which the vibration magnitude exceeded the chosen threshold. To improve robustness and avoid reporting transient artifacts caused by handling or brief impacts, events shorter than 1 s were discarded. The analysis was executed on a background thread to preserve UI responsiveness, and the resulting report

was propagated to the main thread for visualization.

The generated report was presented through a dashboard-style interface conveying a global safety state (safe vs high exposure), summary statistics, and a color-coded event list reflecting event intensity. Although this offline functionality is no longer part of the final application, it played a key role in validating design choices and guiding the development of the real-time pipeline.

**Real-time streaming mode (Movesense BLE + on-device processing)** In the final configuration, the application operates exclusively in real-time mode and connects directly to the Movesense sensor via BLE using the `MovesenseMds` framework (`MDSWrapper`) in combination with CoreBluetooth for device discovery and connection management. The app scans for available peripherals whose name contains `"Movesense"`, connects to the first matching device, and subscribes to the accelerometer stream at 833 Hz through the MDS endpoint `/Meas/Acc/833`. Each received packet contains an `ArrayAcc` field with multiple tri-axial samples, which are parsed and processed sequentially.

The on-device processing pipeline is designed to be computationally lightweight and suitable for real-time execution. Incoming samples are processed sample-by-sample and passed through a band-pass filtering stage implemented using cascaded second-order IIR (biquad) sections. A high-pass cutoff at 20 Hz (applied with two cascaded stages) is used to suppress quasi-static components and low-frequency motion artifacts such as hand movements, while a low-pass cutoff at 400 Hz attenuates high-frequency noise. This band-pass configuration was selected as a pragmatic compromise to isolate frequency components typically associated with power-tool-induced vibrations, while avoiding the complexity of full ISO frequency weighting in a real-time prototype.

Filtered samples are aggregated into fixed-length windows corresponding to approximately 0.5 s of data (416 samples at 833 Hz). For each window, the RMS vibration magnitude is computed and compared against a predefined threshold to classify the window as either `WORK` (vibration exposure) or `REST`. This rule-based decision strategy was chosen for its interpretability, low computational cost, and suitability for the project's research goals.

When a window is classified as `WORK`, the application increments the cumulative exposure time by the window duration and accumulates the squared RMS quantity for exposure estimation. Based on these accumulated quantities, the app computes a running $A(8)$-like indicator by normalizing the average vibration magnitude to an 8-hour reference duration. While this computation does not implement the full ISO 5349 frequency weighting and exposure formalism, it provides a consistent and interpretable proxy for cumulative vibration exposure within the scope of this prototype. Achieving full ISO compliance would require additional frequency weighting, measurement placement considerations, and calibration steps, which are left for future work.

In addition to the instantaneous `WORK/REST` state, the real-time interface displays the current window RMS value, cumulative exposure time, and the running $A(8)$ estimate. A reset action is provided to clear exposure counters and restart accumulation without restarting the application. Acquisition and signal processing operate continuously as samples are received, while UI updates are dispatched on the main thread to ensure responsive interaction.

**UI and interaction** The user interface is designed to reflect the underlying system state in a clear and interpretable manner. In its final form, the real-time interface displays connection status, the current activity classification (`WORK` or `REST`), the short-time RMS vibration level, cumulative exposure time, and the running $A(8)$ indicator. During earlier development stages, the offline interface provided a file-import workflow and a report-oriented view including summary metrics and a detailed event log.

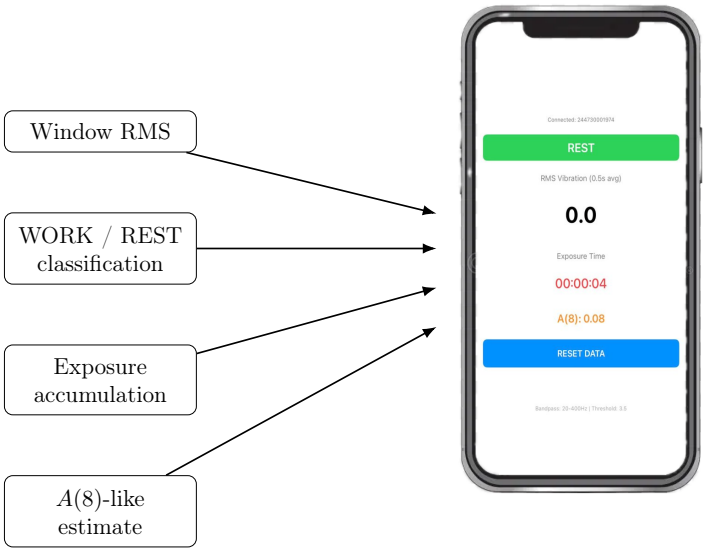



Figure 7: User interface of the real-time mobile application during operation. The interface displays the current activity state (`WORK`/`REST`), short-time RMS vibration level, cumulative exposure time, and the running $A(8)$-like indicator.

# 5 Results

## 5.1 Raw Signal Characteristics

This subsection provides an empirical overview of the raw acceleration signals prior to preprocessing. The focus is on the qualitative differences in vibration morphology and amplitude across the selected power tools.

Figure 5 illustrates a representative tri-axial acceleration segment from an angle grinder. The signal is characterized by high-amplitude, rapid fluctuations with significant peak-to-peak variability. A notable observation is the anisotropic nature of the vibrations; the three axes exhibit distinct intensities, reflecting how vibration energy is transmitted through the tool's chassis to the hand-arm system.

Furthermore, the acceleration magnitude reveals short-term amplitude modulations. These "envelopes" in the raw data likely correspond to physical interactions such as grip force adjustments or changes in tool loading during operation. While sustained activity is common to all tools, their temporal "signatures" differ. High-frequency tools (e.g., the multifunction tool) produce a dense, noise-like trace in the time domain.

Conversely, lower-frequency tools (e.g., the drill) exhibit smoother, more periodic oscillations. These raw recordings also contain significant quasi-static offsets (shifts in the baseline) caused by the orientation of the sensor relative to gravity and the user's arm movements.

To complement this time-domain inspection, frequency-domain analyses were performed to isolate the vibration energy from motion artifacts. Power spectral density (PSD) estimates indicate that tool-induced vibrations are primarily concentrated in frequency bands well above the low-frequency range ($< 10 - 15$ Hz) typically associated with voluntary hand motion. The dominant spectral content varies significantly across the dataset: some tools exhibit broad high-frequency distributions, while others show localized spectral peaks tied to specific mechanical excitation mechanisms. A comprehensive gallery of these spectral profiles for all considered tools is provided in Appendix B. These observations confirm that the raw signals contain rich discriminative information but require targeted filtering to remove the non-informative low-frequency components identified here.

## 5.2 Frequency-Domain Characteristics

To identify the dominant frequency bands associated with tool operation, we analyzed the spectral composition of the signals using both Fast Fourier Transforms (FFT) and Welch's Power Spectral Density (PSD) estimates.

Figure 8 demonstrates that for an angle grinder, the vibration energy is primarily concentrated in the mid-to-high frequency range. Specifically, we observe pronounced spectral content beginning at approximately 20-30 Hz and extending up to several hundred Hz. Distinct peaks appear at frequencies corresponding to the tool's internal rotational dynamics and mechanical resonances.

In contrast, the region below 10 Hz is dominated by handling effects and voluntary hand motion, which are distinct from the tool-induced vibrations. Comparative analysis shows that spectral "signatures" are highly tool-dependent. The multifunction tool, for instance, exhibits a much broader high-frequency distribution than the drill, which displays more localized, narrow-band peaks. These frequency-domain insights directly justify our subsequent preprocessing choices: a high-pass filter is required to suppress the low-frequency motion artifacts identified here, while spectral centroid features will be used in the classification stage to capture these distinct shifts in energy concentration. (For a full comparative library of tool spectra, see Appendix B).

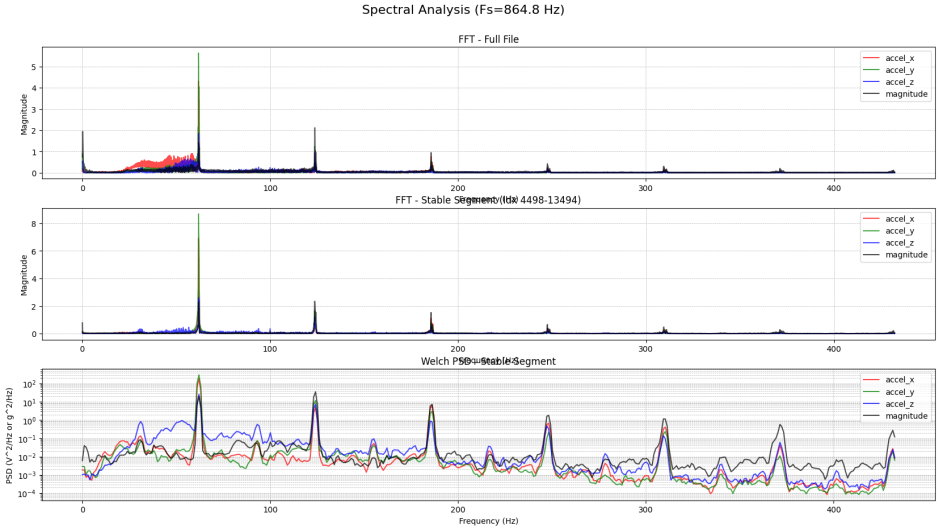



Figure 8: Frequency-domain analysis of the angle grinder recording. The figure reports FFT magnitude spectra (full file and stable segment) and Welch PSD estimates, highlighting low-frequency components, dominant mid-frequency vibration content, and higher-frequency contributions.

### 5.3 Feature Behavior Across Tools

TBC.

### 5.4 Rule-Based Classification Outcomes

TBC.

### 5.5 Exposure Indicators and $A(8)$-like Estimation

TBC.

# 6 Discussion

TBC.

# 7 Conclusion

TBC.

## 8 Contributions

TBC.

# References


[1] R. G. Radwin, T. J. Armstrong, and E. VanBergeijk, "Vibration exposure for selected power hand tools used in automobile assembly," *American Industrial Hygiene Association Journal*, vol. 51, no. 9, pp. 510–518, 1990.

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[10] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*. Pearson, 2010.

# Appendices

## A Swift Code

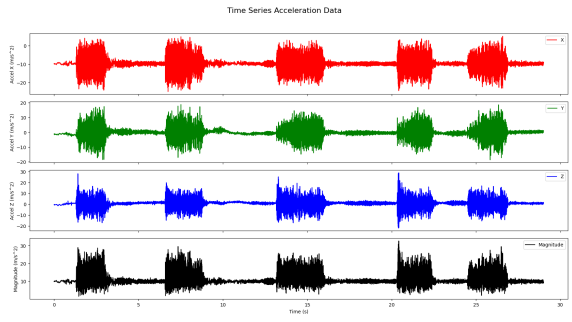
To be completed.

## B Exploratory Signal Analyses

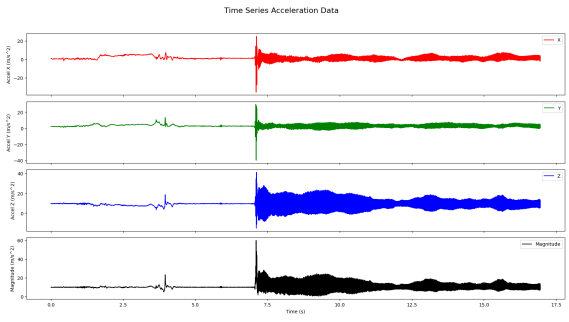
This appendix reports exploratory analyses conducted during the development phase of the project to characterize the vibration signals and inform design choices. These analyses are not part of the real-time processing pipeline described in the Methods section but provide supporting evidence for preprocessing parameters, feature selection, and classification thresholds.

### Multi-Tool Time-Domain Signal Comparison

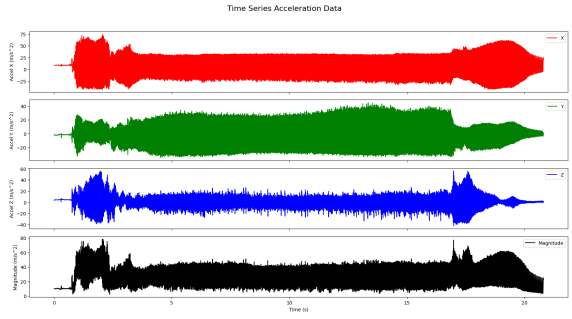
Representative raw acceleration time series for all considered tools are shown in Figure 9. The plots highlight differences in amplitude, oscillatory density, and axis-wise variability across tools.



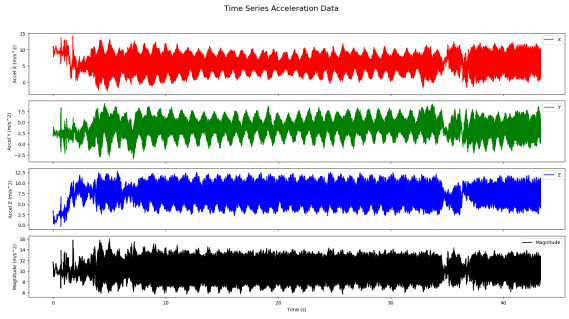

(a) Drill (raw)




(b) Sander (raw)




(c) Angle grinder (raw)

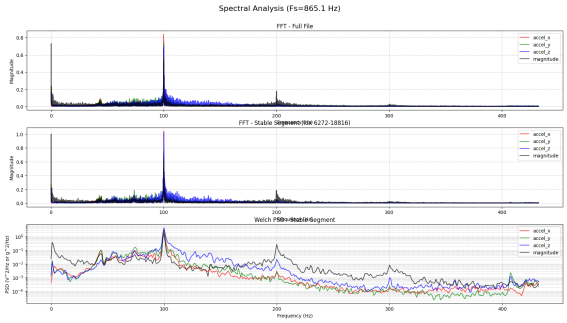



(d) Multifunction tool (raw)

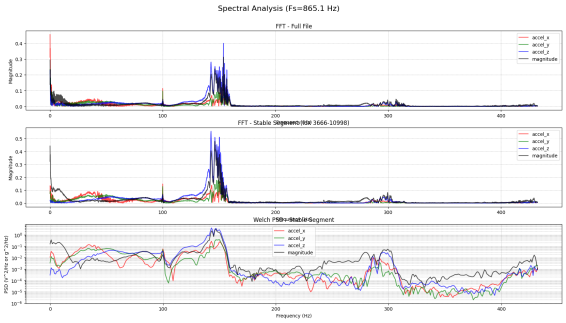
Figure 9: Raw tri-axial acceleration time series for the four tools, shown with axis components and magnitude. These plots provide a qualitative comparison of amplitude variability and oscillatory density across tools prior to preprocessing.

### Frequency-Domain Analysis Across Tools

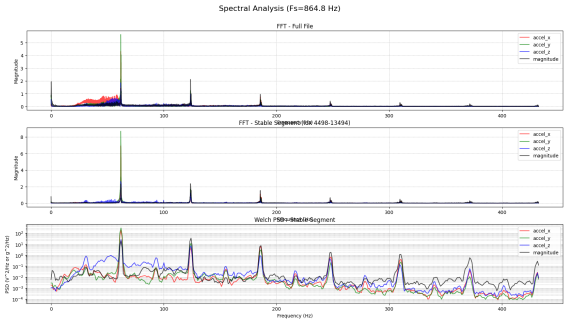
Frequency-domain representations (FFT and Welch PSD) were computed for representative stable segments of each tool recording. Figure 10 illustrates the diversity of spectral signatures observed across tools, including differences in dominant frequency bands and bandwidth.



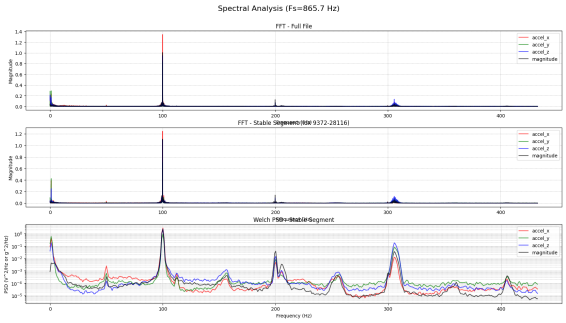

(a) Drill




(b) Sander




(c) Angle grinder




(d) Multifunction tool

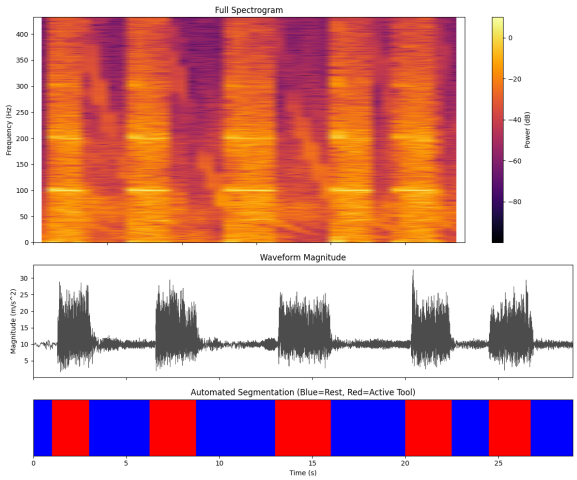
Figure 10: FFT and Welch PSD analyses for representative stable segments of each tool. The plots highlight tool-dependent spectral signatures, including dominant frequency bands and narrow peaks superimposed on broadband vibration content.

## Feature Space Visualization and Threshold Inspection

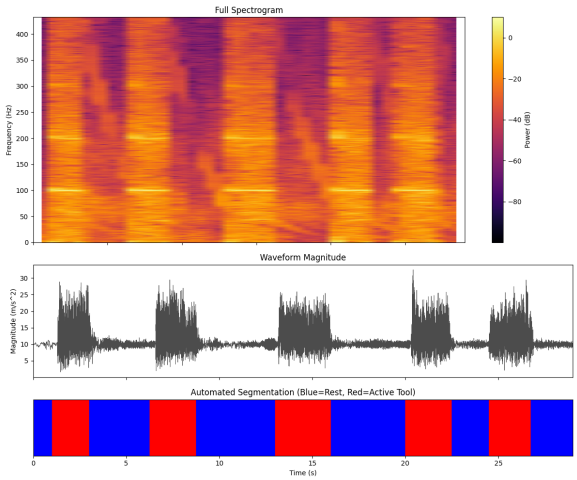
To support the selection of window-level features and classification thresholds, the distribution of RMS vibration magnitude and spectral centroid values was analyzed. Figure 11 shows feature-space visualizations highlighting separability between tool operation and non-tool activity.

## Advanced Decomposition Analysis

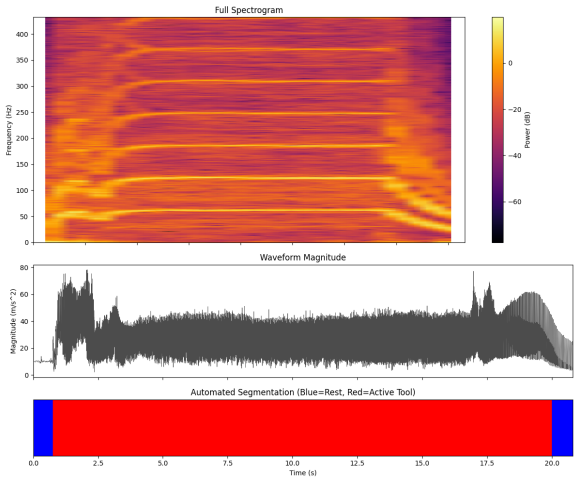
To be completed.



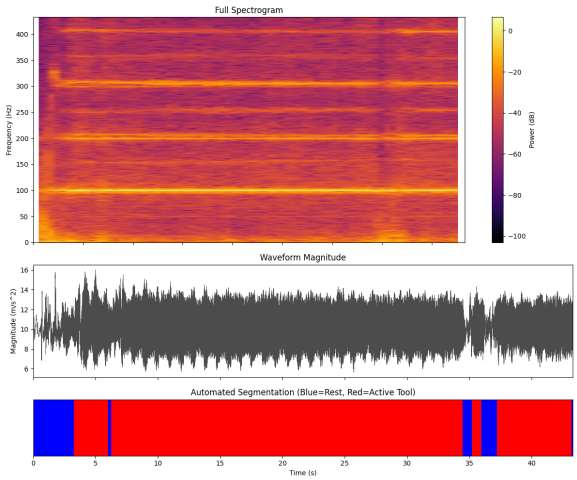

(a) Drill




(b) Sander




(c) Angle grinder




(d) Multifunction tool

Figure 11: Illustrative window-level feature trajectories and rule-based classification outputs for each tool. The figures show (top) filtered signal magnitude, (middle) window classification timeline, and (bottom) RMS and spectral centroid values compared against the selected thresholds. **Note**: Sander plot is not correct, we will fix it for the final report.